

JIAJUN (NICK) HUO

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SUMMARY

AI & backend engineer (MS IS @ UIUC) with 2+ years shipping multi-agent systems, GraphRAG, and data infra that cut costs and scale 10x. Want to spend the next few years building agentic AI that is reliable enough for real, high-stakes products

EDUCATION

University of Illinois Urbana Champaign, M.S. in Information Systems Aug 2024 – May 2026

GPA 3.86/4.0 · Distributed Systems & Database Systems & Applied Machine Learning · U.S. News #1 Information Systems

Shenzhen University, B.S. in Mathematics Sep 2019 – Jun 2023

GPA 3.79/4.50 (Top 10 %) · 2x merit scholarships · 4 national awards in ML modeling (supply-chain, finance, computer-vision)

WORK EXPERIENCE

Beckman Institute for Advanced Science and Technology Mar 2026 – Present

AI Engineer & Graduate Researcher Urbana, IL

- Architected multi-agent system with tool schemas via OpenAI function-calling on a Neo4j-backed closed-loop pipeline
- Designed preference learning with per-user SGD updates from explicit feedback, driving personalized-aware reranking
- Built a graph RAG pipeline: LLM-extracted typed edges, cycle-detection validation, prereq-chain subgraph retrieval
- Engineered persistent learner memory with cross-session mastery state, grounding generation on evolving user knowledge

H60 Consulting (via UIUC Business Intelligence Group) Aug 2025 – May 2026

Software Engineer & Capstone Project Lead Remote · Chicago HQ

- Led 10-person team for flagship RAG copilot, transforming vague visions into a benchmark project for a top global airline
- Designed Claude Haiku query-rewrite layer translating pilot phrasing into doc-aligned vocab; improved 94% test queries
- Shipped Pinecone hybrid retrieval (BM25+dense, custom tokenizer) lifting top-5 hit rate from 84.3% to 88.1%
- Built RAG eval (RAGAS + deterministic citation/coverage/refusal layer); closed-loop tuning lifted citation accuracy 60%

Donut Labs Jun 2025 – Aug 2025

Software Engineer Intern – Sequoia-backed crypto agent for pro trader Remote · NYC HQ

- Governed MCP tool-use across 46 actions with enforced OpenAPI contracts and rollout guards, raising task success 22%
- Engineered a context reuse strategy with Redis caching for model intermediate states, cutting GPU costs by \$2K/month
- Built a self-improving RAG pipeline with data flywheel on LanceDB, decreasing real-time per-tweet analysis costs by 86%

Sonic SVM Mar 2023 – Jul 2024

Software Engineer, Data Infra – Bitkraft-backed SaaS for game studios Shenzhen

- Launched a self-service analytics platform on GCP with Looker, enabling product teams to answer 80% of data questions
- Scaled data ingestion 10x using Kafka & Spark to support 200% user growth and international product expansion
- Built monitoring with Prometheus & Grafana, reducing critical data incident response time from hours to under 5 minutes

Baidu Apr 2022 – Jul 2022

Software Engineer, ML Intern – Premier AdTech Platform with 10,000+ Global Partners Shenzhen

- Drove 3x ad revenue growth to \$200K/month in 3 months by optimizing traffic arbitrage for emerging publisher clients
- Conceived 4+ AB testings across user lifecycle to supercharge AARRR metrics, boosting daily new user 1.5x in 1 month

Tencent Nov 2020 – May 2021

Technical Product Manager Intern – The largest WeChat ecosystem platform for universities Shenzhen

- Drove the full SDLC for campus app, mitigating risks and unblocking dependencies to launch from scratch in 12 weeks
- Translated product requirements into technical specifications for a real-time notification system serving 50k+ students

SKILLS

Languages & Frameworks Python, TypeScript/JavaScript, Go, SQL, Rust, Next.js, Tailwind, HTML, CSS, Bash/Shell
Infrastructure & Data AWS (EC2/S3/Lambda), Docker, Kubernetes, Terraform, PostgreSQL, Redis, MongoDB